

COLUMBUS RICKENBACKER, OH, USA

WMO: 724285

Lat: 39.817N

Lon: 82.933W

Elev: 227

StdP: 98.63

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 13812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-12.2	-19.8	0.7	-13.4	-17.0	0.9	-11.3	12.1	2.3	11.1	1.5	2.7	270	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	33.4	23.3	32.3	22.9	31.1	22.4	26.5	30.0	25.3	29.8	24.2	28.1	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.5	28.9	23.9	19.3	27.2	22.9	18.1	26.2	84.1	30.1	78.3	29.6	74.5	27.2	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	8.9	7.9	-19.1	35.6	4.3	3.5	-22.2	38.1	-24.7	40.2	-27.1	42.2	-30.3	44.8		
(5)			-19.5	27.2	4.2	1.9	-22.5	28.5	-24.9	29.7	-27.3	30.8	-30.3	32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	-0.9	0.9	5.9	12.4	18.1	22.7	24.3	23.6	20.0	13.2	6.6	1.7
	DBStd	10.21	6.90	6.21	6.11	5.03	4.56	3.31	2.66	2.72	3.80	4.90	5.16	5.61
	HDD10.0	1208	343	261	159	32	2	0	0	0	0	22	127	263
	HDD18.3	2794	596	489	388	187	63	6	0	1	26	172	352	516
	CDD10.0	2094	5	6	32	103	252	381	443	421	300	121	26	6
	CDD18.3	640	0	0	2	8	56	137	184	163	76	13	0	0
	CDH23.3	5730	0	1	19	88	514	1213	1672	1424	674	123	2	0
	CDH26.7	2073	0	0	3	14	162	455	657	526	229	28	0	0
Wind	WSAvg	3.3	4.0	3.8	3.9	4.0	3.3	2.9	2.5	2.3	2.5	3.1	3.4	3.6
Precipitation	PrecAvg	1025	74	57	87	98	107	105	109	86	84	70	67	79
	PrecMax	1345	217	117	193	189	201	231	265	195	154	161	126	183
	PrecMin	732	27	25	42	33	29	24	46	26	27	20	19	32
	PrecStd	144	38	23	34	33	48	45	44	42	36	33	28	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.7	26.1	29.1	32.4	34.0	35.5	34.9	33.9	30.0	22.5	18.0
		MCWB	15.5	14.9	17.0	18.3	20.5	23.2	24.4	23.6	21.3	20.4	15.8	15.4
	2%	DB	15.0	16.1	22.1	26.3	30.4	32.5	33.1	32.8	31.3	27.0	19.9	15.1
		MCWB	13.0	11.9	15.2	16.9	20.4	22.8	24.1	23.1	21.7	19.2	14.5	12.8
	5%	DB	12.4	12.9	18.6	23.7	28.6	31.2	32.2	31.3	29.1	24.0	17.4	12.6
		MCWB	10.7	10.3	13.4	15.8	20.2	22.3	23.6	22.8	20.9	17.7	13.8	10.6
	10%	DB	8.7	9.9	15.9	21.3	26.4	29.3	30.7	29.7	27.4	21.6	15.1	10.1
		MCWB	7.0	7.1	11.6	14.7	19.0	21.3	22.9	22.0	20.0	16.7	11.6	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	16.0	18.0	21.7	24.8	26.9	28.3	27.8	25.3	22.4	18.2	16.2
		MCDB	17.4	19.2	23.1	26.0	28.8	30.7	32.1	31.3	29.3	27.2	20.8	17.8
	2%	WB	13.6	13.1	16.0	18.9	22.5	25.3	26.6	26.1	23.7	20.7	16.0	13.1
		MCDB	15.1	14.3	20.8	22.6	26.8	30.2	30.4	29.5	28.0	24.3	18.6	14.8
	5%	WB	10.7	10.5	14.4	17.2	21.2	24.0	25.4	24.9	22.6	19.2	14.2	11.1
		MCDB	12.0	12.9	17.8	20.8	25.8	28.2	29.9	28.9	26.8	22.6	16.6	12.4
	10%	WB	7.2	7.5	12.2	15.7	20.1	23.0	24.3	23.8	21.5	17.6	12.1	8.0
		MCDB	8.5	9.3	15.3	20.1	25.0	27.0	28.0	27.2	25.3	20.6	14.8	9.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	10.1	11.9	11.3	10.8	10.7	11.1	11.9	11.2	9.7	7.5
		MCDBR	10.4	13.0	14.5	14.7	13.7	12.8	12.3	12.9	14.4	14.2	13.2	10.5
		MCWBR	8.8	10.1	9.0	8.1	6.1	5.2	5.0	5.1	5.8	7.5	9.0	8.8
	5% WB	MCDBR	9.8	12.3	12.2	12.2	11.7	10.9	10.8	10.5	11.4	11.5	11.0	10.2
		MCWBR	8.8	10.5	8.6	8.2	6.7	5.5	5.7	5.5	5.6	6.9	8.7	9.4
Clear-Sky Solar Irradiance	taub	0.312	0.331	0.360	0.397	0.441	0.458	0.467	0.460	0.400	0.366	0.329	0.307	
	taud	2.422	2.344	2.336	2.258	2.169	2.166	2.152	2.152	2.299	2.386	2.454	2.456	
	Ebn at Noon	852	883	892	878	843	826	814	809	839	832	825	830	
	Edh at Noon	82	102	115	132	148	148	149	146	118	97	79	74	
All-Sky Solar Radiation	RadAvg	1.66	2.49	3.51	4.74	5.48	6.01	5.83	5.43	4.43	3.08	2.09	1.41	
	RadStd	0.16	0.32	0.32	0.41	0.46	0.41	0.33	0.29	0.41	0.28	0.27	0.16	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (37 neighbors)		N/A	N/A	-2.72	N/A	-1.65	-2.20	N/A	N/A	N/A
		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	+129	+71

Nomenclature: See separate page